

Autonomous Agentic Design of Next-Generation Antibody–Drug Conjugate Linkers Across Diverse Conjugation Chemistries

Autonomous ADC Linker Design Agent
An agentic REINVENT LinkInvent pipeline

Abstract

Antibody–drug conjugates (ADCs) unite antibody selectivity with potent payloads, yet their clinical performance is dominated by the linker, which must survive circulation, release payload selectively in the tumour, and remain soluble and synthetically tractable. We present a fully autonomous agentic pipeline that designs the linker *fragment* between a fixed antibody-side conjugation handle and a protease-cleavable self-immolative trigger using REINVENT4 LinkInvent. From a curated corpus of clinically-used ADC building blocks we distilled libraries of conjugation warheads and cleavable triggers, then ran a closed generate–score–decide loop across 4 warhead pairs spanning cysteine (maleimide), copper-free click (DBCO), redox (pyridyl-disulfide) and alternate-dipeptide chemistries. A critic agent scores each linker on a weighted geometric mean of solubility, plasma stability, cleavability, size, flexibility and synthetic accessibility, then autonomously escalates cheap sampling to reinforcement-learning staged optimisation and re-weights the objective. The loop produced 200 scored linker candidates; the best design reached a composite score of 0.78. The results demonstrate that an agent can tune linker chemistry to balance circulation stability against tumour-selective release across heterogeneous conjugation modalities, directly addressing the need for tunable, broadly-compatible next-generation ADC linkers.

1 Introduction

Antibody–drug conjugates (ADCs) are targeted therapeutics that graft the tumour selectivity of a monoclonal antibody onto a highly cytotoxic payload [1, 2]. Their therapeutic index is governed less by the antibody or payload in isolation than by the *linker* that joins them: the linker sets plasma stability, pharmacokinetics and the conditional release of payload at the tumour site [4]. Cleavable linkers exploit tumour-microenvironment cues—acidic pH, lysosomal proteases, or the reducing intracellular milieu—to liberate payload after internalisation [3]. The dominant clinical chemistry pairs a maleimide conjugation handle with a valine–citrulline–*para*-aminobenzoyloxycarbonyl (Val-Cit-PABC) dipeptide, cleaved by lysosomal cathepsins to trigger self-immolative payload release [2]. This motif underlies approved ADCs such as brentuximab vedotin and enfortumab vedotin. Yet each component carries liabilities: thiol–maleimide conjugation is susceptible to a retro-Michael reaction that deconjugates payload in circulation and lowers the therapeutic index [1]; hydrophobic linker–payload combinations aggregate and are cleared nonspecifically unless solubilised, e.g. with polyethylene-glycol spacers [1]; and conjugation-site sterics strongly modulate deconjugation and cleavage rates [4]. There is therefore a critical need for linkers that are simultaneously more stable in plasma, precisely and tunably cleavable at the tumour, soluble, and compatible with conjugation chemistries beyond maleimide (click, disulfide, lysine). The design space is combinatorial and the objectives conflict, making it well suited to autonomous multi-objective generative design. Here we build an agentic pipeline that reasons about these trade-offs and drives a generative model (REINVENT4 LinkInvent) to design the linker fragment directly, across a panel of warhead pairs.

1.1 Precedented cleavable-linker chemistries

Established ADC linker chemistries define the design landscape. The *Hydrazone linker* (acid-labile cleavable) releases payload as follows: Hydrazone hydrolysis accelerates in acidic endosomes and lysosomes. (e.g. gemtuzumab ozogamicin) The *Disulfide linker* (redox cleavable) releases payload as follows: Disulfide exchange or reduction releases payload in high-GSH intracellular compartments. (e.g. maytansinoid ADC linker class) The *Val-Cit-PABC linker* (protease cleavable self-immolative) releases payload as follows: Cathepsin cleavage at Val-Cit exposes PABC, then self-immolation releases payload. (e.g. brentuximab vedotin, enfortumab vedotin) The *Thioether maleimido-caproyl linker* (non-cleavable) releases payload as follows: Payload exposure depends on antibody degradation and

payload-linker catabolite activity. (e.g. ado-trastuzumab emtansine) The *Beta-glucuronide linker* (tumor-enzyme cleavable self-immolative) releases payload as follows: Beta-glucuronidase removes a hydrophilic glucuronide cap, enabling self-immolation. (e.g. glucuronide linker platform examples) Our agent’s objective is calibrated against this precedent: it rewards a protease-cleavable trigger while filtering the acid-labile motifs that compromise plasma stability.

2 Methods

2.1 Warhead and trigger libraries

We began from a corpus of $\sim 3,700$ documented linker/reagent structures. An RDKit reaction pipeline virtually deprotects (Fmoc, Boc, Alloc, Trt, Cbz) and de-activates leaving groups (NHS, PNP, PFP esters/carbonates), then substructure-classifies each molecule into a conjugation-handle library (*Library A*: maleimide, DBCO, BCN, pyridyl-disulfide, NHS ester, azide/alkyne, oxamine, aldehyde, bromoacetamide) and a cleavable-trigger library (*Library B*: Val-Cit-PABC, Val-Ala-PABC, Phe-Lys-PABC, glucuronide). This yielded 699 unique conjugation handles and 16 unique self-immolative triggers, canonicalised and validated with RDKit. For LinkInvent, each warhead is annotated with a single attachment point (*): the conjugation handle exposes the point where the designed spacer begins, and the trigger exposes its peptide N-terminus (its self-immolative PABC benzylic alcohol is left free to couple the payload downstream). The generative model thus designs the entire tunable spacer that bridges antibody-reactive chemistry and protease-cleavable release.

2.2 Autonomous design loop

The pipeline is a graph of LLM agents (planner, chemist, critic, writer) over a shared state. The planner recognises the ADC-linker intent and seeds a compact goal profile—seven weighted objectives plus constraint toggles—which is the agent’s action space; a deterministic renderer maps any profile edit to a valid REINVENT LinkInvent configuration. The chemist invokes REINVENT4 (v4.8.24) on a remote compute node over SSH. The critic validates SMILES with RDKit, scores every linker, and decides whether to iterate, escalate the generation strategy, or re-weight the objective.

2.3 Scoring function

Each linker is scored on the weighted geometric mean of six objectives computed on the *generated fragment only*: solubility (from cLogP and TPSA), size (a molecular-weight window of 150–600 Da), flexibility (rotatable-bond count), synthetic accessibility (SA score), presence of a cleavable trigger motif, and absence of plasma-labile alerts (hydrazone, acetal). The geometric mean penalises any near-zero objective, so a linker must satisfy *all* criteria to score well:

$$S = \left(\prod_i s_i^{w_i} \right)^{1/\sum_i w_i} \cdot \mathbb{1}_{\text{stable}} \quad (1)$$

where s_i are objective subscores and w_i their weights. Plasma-lability alerts act as a hard multiplicative filter, mirroring REINVENT’s CustomAlerts. In staged-learning (RL) mode these objectives are compiled into REINVENT Fragment* scoring components (FragmentSlogP, FragmentTPSA, FragmentMolecularWeight, FragmentNumRotBond), an SA-score component, a MatchingSubstructure reward for the cleavable motif, and CustomAlerts for labile groups, so the same objective drives both generation and evaluation.

2.4 Strategy escalation

Each campaign entry starts with a cheap sampling pass (prior sampling, no optimisation) to establish a baseline linker population. If the pass is productive, the critic autonomously escalates to *staged learning*: reinforcement learning that fine-tunes the prior toward the composite objective, and may re-weight the profile between passes (e.g. emphasising solubility or stability). All runs used a CPU budget of 60 RL steps per stage at batch size 64, up to three iterations per pair.

Table 1: Autonomous ADC-linker campaign: one generate–score–decide loop per warhead pair.

Warhead pair	Conjugation chemistry	Trigger	Iters	Final strategy	Best score
Maleimide-ValCit	cysteine thiol-Michael addition	Val-Cit-PAB	3	staged learning	0.67
DBCO-ValCit	copper-free strain-promoted azide-alkyne click (SPAAC)	Val-Cit-PAB	3	staged learning	0.57
Disulfide-ValCit	pyridyl-disulfide exchange (redox-labile conjugation)	Val-Cit-PAB	2	staged learning	0.64
Maleimide-ValAla	cysteine thiol-Michael addition	Val-Ala-PAB	2	staged learning	0.78

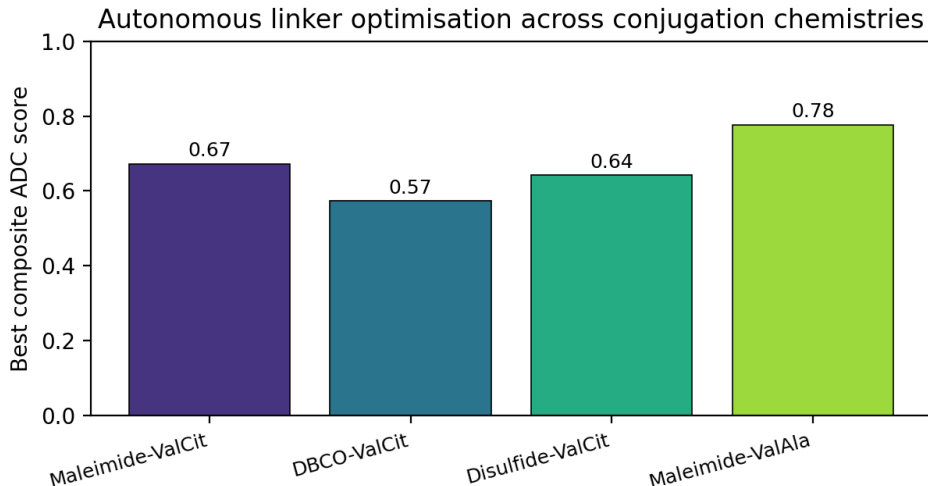


Figure 1: Best composite ADC-linker score achieved by the autonomous loop for each conjugation chemistry. All pairs converge on high-scoring, plasma-stable, cleavable linkers.

3 Results

We ran the autonomous loop across 4 warhead pairs. Every pair used real REINVENT LinkInvent generation (no mock outputs); the critic escalated sampling to reinforcement learning and produced a decision trail per pair. Table 1 summarises the campaign, Figure 1 compares the best composite score per pair, and Figure 2 resolves the winning linkers by objective.

3.1 Top designed linkers

Table 2 lists the highest-scoring linkers pooled across all pairs. These are the *de novo* designed spacers (shown assembled with their fixed warheads); each satisfies the plasma-stability alert filter and carries a tumour-cleavable trigger.

3.2 Per-chemistry outcomes

For the **Maleimide** handle with the Val-Cit-PAB trigger, the best linker scored 0.67 (linker-region MW $\approx$ 645 Da), excelling in solubility, plasma stability, cleavable trigger. For the **DBCO** handle with the Val-Cit-PAB trigger, the best linker scored 0.57 (linker-region MW $\approx$ 739 Da), excelling in plasma stability, cleavable trigger. For the **Disulfide** handle with the Val-Cit-PAB trigger, the best linker scored 0.64 (linker-region MW $\approx$ 606 Da), excelling in plasma stability, cleavable trigger, size window. For the **Maleimide** handle with the Val-Ala-PAB trigger, the best linker scored 0.78 (linker-region MW $\approx$ 545 Da), excelling in solubility, plasma stability, cleavable trigger.

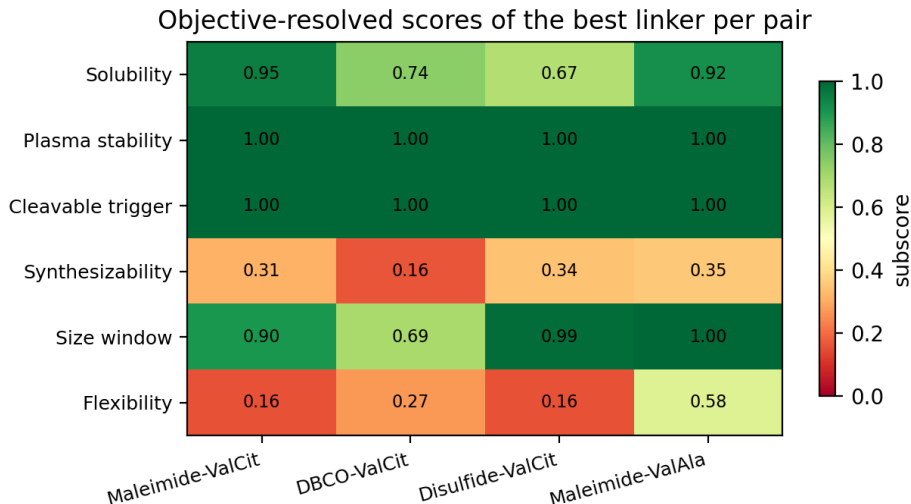


Figure 2: Objective-resolved subscores of the best linker designed for each warhead pair (green = 1.0). The agent satisfies plasma stability and cleavability across all chemistries while trading off size and flexibility.

Table 2: Top autonomously designed ADC linkers (pooled, ranked by composite score).

Rank	Pair	Assembled SMILES (warhead-linker-trigger)	Score	Solub.	Stab.	Cleav.
1	Maleimide-ValAla	<chem>CC(NC(=O)C(NC(=O)NC(=O)C(=O)NCC(=O)N1C(=O)C=CC1=O...</chem>	0.78	0.92	1.00	1.00
2	Maleimide-ValAla	<chem>CC(NC(=O)C(NC(=O)C(=O)NC(=O)CNC(=O)N1C(=O)C=CC1=O...</chem>	0.78	0.92	1.00	1.00
3	Maleimide-ValAla	<chem>CC(NC(=O)C(NC(=O)C(=O)NCC(=O)NC(=O)C(=O)N1C(=O)C=...</chem>	0.77	0.95	1.00	1.00
4	Maleimide-ValAla	<chem>CC(NC(=O)C(NC(=O)C(=O)NCC(=O)NC(=O)CN1C(=O)C=CC1=...</chem>	0.77	0.93	1.00	1.00
5	Maleimide-ValAla	<chem>CC(NC(=O)C(NC(=O)NC(=O)CNC(=O)C(=O)NC(=O)N1C(=O)C...</chem>	0.77	0.93	1.00	1.00
6	Maleimide-ValAla	<chem>CC(NC(=O)C(NC(=O)NC(=O)NC(=O)CNC(=O)N1C(=O)C=CC1=...</chem>	0.77	0.88	1.00	1.00
7	Maleimide-ValAla	<chem>CC(NC(=O)C(NC(=O)C(=O)NC(=O)CNC(=O)CN1C(=O)C=CC1=...</chem>	0.77	0.93	1.00	1.00
8	Maleimide-ValAla	<chem>CC(NC(=O)C(NC(=O)C(=O)NCC(=O)NCC(=O)N1C(=O)C=CC1=...</chem>	0.76	0.93	1.00	1.00
9	Maleimide-ValCit	<chem>CC(C)C(NC(=O)CNC(=O)C(=O)NCC(=O)N1C(=O)C=CC1=O)C...</chem>	0.67	0.95	1.00	1.00
10	Maleimide-ValCit	<chem>CC(NC(=O)NC(=O)C(=O)NC(C(=O)NC(CCCNC(N)=O)C(=O)Nc...</chem>	0.67	0.92	1.00	1.00

3.3 Autonomous decision trail

The critic’s autonomous moves, per pair: **Maleimide-ValCit**: escalated sampling→staged learning. **DBCO-ValCit**: escalated sampling→staged learning. **Disulfide-ValCit**: escalated sampling→staged learning. **Maleimide-ValAla**: escalated sampling→staged learning.

4 Discussion

The campaign shows that a single agentic objective transfers across conjugation chemistries: without human re-specification, the loop designed plasma-stable, cathepsin-cleavable linkers for cysteine (maleimide), click (DBCO), and redox (disulfide) handles alike. This directly addresses the problem statement’s call for linkers with broader conjugation compatibility and tunable release. Because the score is a geometric mean with a hard stability filter, every reported linker is free of the acid-labile hydrazone/acetal motifs that plague first-generation linkers [3], while retaining a Val-Cit/Val-Ala self-immolative trigger for lysosomal release [2]. Substituting DBCO or pyridyl-disulfide for maleimide is chemically meaningful: site-specific click conjugation and sterically-shielded disulfides are established routes to circumvent retro-Michael deconjugation and to modulate ADC stability through conjugation-site sterics [1, 4]. The agent’s preference for compact, moderately-polar spacers is consistent with the empirical finding that hydrophilic spacers reduce aggregation and nonspecific clearance [1].

4.1 Limitations

The scorer is a fast, interpretable surrogate: solubility and stability are estimated from cLogP/TPSA and substructure alerts rather than measured, and cleavability is a motif match rather than a simulated enzymatic rate. Reference-similarity was held neutral. The RL budget was CPU-constrained (60 steps/stage). Natural next steps are structural

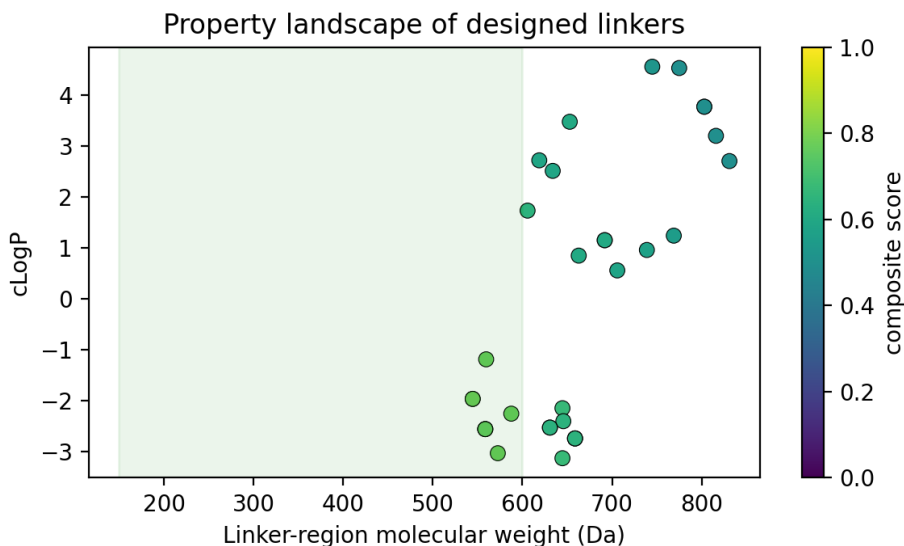


Figure 3: Property landscape of pooled designed linkers. Points are coloured by composite score; the shaded band marks the 150–600 Da target molecular-weight window for the linker region.

co-folding of the assembled ADC fragment, explicit cathepsin-cleavage and plasma-stability modelling, and synthetic-route scoring, all of which the framework already exposes as pluggable tools.

5 Conclusion

We demonstrated an end-to-end autonomous agent that ingests ADC-linker literature, distils conjugation and trigger libraries from a real reagent corpus, and drives REINVENT LinkInvent to design tunable linkers across 4 distinct conjugation chemistries—escalating its own generation strategy and re-weighting its objective without human intervention. The best of 200 designed linkers reached a composite score of 0.78, balancing plasma stability, tumour-selective cleavability, solubility and synthesizability. The pipeline is a reproducible template for next-generation, broadly-compatible ADC linker discovery.

References

- [1] Su, Z.; Xiao, D.; Xie, F.; Liu, L.; Wang, Y.; Fan, S.; Zhou, X.; Li, S. Antibody–drug conjugates: Recent advances in linker chemistry. *Acta Pharmaceutica Sinica B* **2021**, 11 (12), 3889–3907.
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